Context Poisoning in LLM Agent Systems

A Systematization of Knowledge — with reconstructed transfer experiments

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1,158 papers · 237 named attacks · 534 defenses · 16 reconstructed defense × attack cells

Two literatures, one failure mode

An agent acts on whatever its context says. Two research communities study that going wrong — separately.

- Security track treats it as an attack — indirect prompt injection, PoisonedRAG, BadRAG, AgentPoison. An adversary writes the context.
- ML/AI track treats the same corruption as an accident — lost-in-the-middle, context rot, knowledge conflict, distraction. Nobody writes it on purpose.
- Both sides build defenses. Each set is evaluated on its own side's failures, and almost never on the other's.
- **The SoK question: is that separation justified by the mechanisms, or is it an artefact of how the two communities publish?**

CORPUS COMPOSITION

626
Security-track papers

512
ML/AI-track papers

20
papers coded as both

How the corpus was built, and how we measure

Every number in this deck is regenerated from data/paper_explorer.db, not transcribed by hand.

PIPELINE

- ~26,000 papers discovered by keyword and citation expansion
- 1,158 curated after screening against explicit criteria
- Registries extracted: 237 attacks, 534 defenses, 349 reported pairs
- Taxonomy cube: 9 channels × 3 intents × 6 consequences

THREE DISCIPLINES, APPLIED MECHANICALLY

- Control fires first — a defense that cannot reproduce its own benefit on its own benchmark yields no transfer verdict.
- Minimum detectable change — two-proportion power analysis (α=.05, power=.80). A smaller delta is not a result, whatever its sign.
- Authors' scorers, prompts and payloads imported unchanged. Harness adaptation is allowed; evaluation rubrics are not touched.

The first two rules are enforced inside the database loader, so the prose cannot drift away from the data. Applying them consistently has already forced two corrections to our own published claims.

The two communities do not read each other

Full-population counts resolved from the citation graph — not a hand-coded sample, so no confidence interval is reported.

Measure	Value	Reading
Taxonomy cube cells that are empty	**98 of 162 (60.5%)**	Most channel × intent × consequence combinations have never been studied
Security papers citing ≥1 ML/AI paper	**70 of 599 (11.7%)**	Nine in ten ignore the incidental literature entirely
ML/AI papers citing ≥1 security paper	**37 of 407 (9.1%)**	The asymmetry is small — both directions are near-silent
Reported pairs staying inside one literature	**341 of 348 (98.0%)**	Only 7 pairs in the entire corpus cross the divide

Why this matters for a security venue. If the two sides never test each other's mechanisms, every claim of the form “this defense is robust” is scoped to one literature's threat model — and nobody has checked whether it survives the other's.

Most defenses have never met most attacks

The coverage matrix is far sparser than the plan anticipated — and the shape of the sparsity is itself the finding.

119
of 237 attacks (50.2%) have no defense pair at all

320
of 534 defenses (59.9%) have no attack pair at all

15
of 534 defenses (2.8%) were validated on both threat models

349
defense × attack pairs reported anywhere

A defense paper with no attack pair is not a neutral gap — a defense is supposed to defend something. Classifying all 320:

Classification of the 320 zero-pair defenses	n	What it means
Non-adversarial by construction	126	Reliability or efficiency work; no threat model is claimed
Names a registry attack but never tests it	**99**	Candidate missed pairs — the strongest lead for new coverage
Threat named in prose only	41	An adversary is described; no experiment follows
Self-constructed threat	22	Evaluated only against an attack the authors invented
Benchmark named, no mechanism	11	Reports a score, not a defense outcome
Unclassified	21	Insufficient extracted text to decide

We ran the code instead of quoting the papers

Every cell below is a defense × attack pair reported nowhere in the literature. That is why it was chosen.

- A reconstructability screen ran first, over 7 candidate attacks, before any GPU time was booked — 2 cleanly runnable, 2 needing substitution, 1 unreadable, 1 miscoded in our own registry.
- 4 attack mechanisms reconstructed: corpus poisoning (signal removal), PoisonedRAG (lie insertion), BadRAG DoS (refusal induction), BadRAG Selective-Fact (tone steering).
- 5 defense implementations run from released code: ParamMute, CK-PLUG, SpARE, RobustRAG KeywordAgg, RobustRAG DecodingAgg.
- Attack dose is swept from 0 to 10 poisoned passages out of 10, so every cell is a curve rather than a single point.

16
defense × attack × model cells

94
dose-level measured conditions

2
defense families compared

3
models, incl. a controlled rerun

Finding 1 — the collapse is family-specific

At full saturation, when every retrieved passage is attacker-controlled, the two families behave categorically differently.

Defense family	Mechanism	Cells with usable control	Resolvable accuracy loss
Context-reliance	Suppress parametric memory so the model follows retrieved text (ParamMute, CK-PLUG, SpARE)	5 of 8	**4**
Isolate-then-aggregate	Answer each passage alone, then combine (RobustRAG KeywordAgg, RobustRAG DecodingAgg)	5 of 5	**0**

- The penalty is invariant: −31.7 and −31.8 pp across two defenses, two attacks and two models — one of them not even instruction-tuned.
- Against lie insertion the harm starts at 8 of 10 poisoned passages, not at 10 of 10. A context-reliance defense is a dose multiplier: every poisoned passage is one more passage it insists on.
- **Design rule — do not suppress parametric memory. It is the model's only recourse when the evidence set empties, and it is worth about 32 points.**

Full curves, controls and limits: family_dissociation.md · saturation_invariant.md · collapse_boundary.md

Finding 2 — some defenses create the vulnerability

Attack success against an undefended model, and against the same model with the defense switched on.

Defense	Attack	Undefended	Defended	Effect of turning the defense on
CK-PLUG	Tone steering	0.0 %	**99.0 %**	78.2 % from a single poisoned passage in ten
ParamMute	Tone steering	0.0 %	**28.8 %**	Same direction, weaker
RobustRAG KeywordAgg on Llama-3-8B-Instruct	Refusal induction	0.0 %	**85.0 %**	The model never refuses until the defense is added
RobustRAG KeywordAgg on Mistral-7B-Instruct	Refusal induction	0.0 %	**5.7 %**	Same defense, same attack, different model — no effect

- **The attack does nothing to an undefended model. It only works through the defense.**
- Why: tone steering does not corrupt the answer, so a defense that pushes the model to follow the retrieved text follows the slant too — and scores itself as working perfectly. The attack and the defense want the same thing.
- The last row is the uncomfortable one: the same defense is protective on one model and harmful on another, so model choice is part of the threat model.

Full curves, controls and limits: family_dissociation.md · second_implementation.md · badrag_transfer_result.md

Finding 3 — two implementations, opposite verdicts

RobustRAG ships two aggregation variants: same isolation step, same prompt builder, different combination rule.

Poisoned passages (of 10)	1	5	6	7	8	9	10
KeywordAgg Δ accuracy	+7.3	+4.0	+4.0	+5.7	+6.7	**+12.3**	−5.7
DecodingAgg Δ accuracy	+8.7	−9.0	−14.0	−18.3	−20.3	**−17.3**	−4.0
Gap between the two	**1.4**	**13.0**	**18.0**	**24.0**	**27.0**	**29.6**	**1.7**

- At 9 of 10 poisoned, one implementation of the same principle is resolvably protective and the other resolvably harmful — 29.6 points apart, both clearing threshold.
- The mechanism was written down and committed before the test: a hard keyword filter discards a passage carrying no answer content, so one honest passage still carries the answer; a soft logit average lets nine junk distributions dilute it.
- **That predicts the gap scales with the honest minority and vanishes when it reaches zero. Measured afterwards: monotone across six doses, then collapsing to 1.7 at saturation. Confirmed.**

Full curves, controls and limits: aggregation_divergence.md · second_implementation.md · CELL_CAMPAIGN.md

Positioning for an A* security venue

Target venues: IEEE S&P · USENIX Security · ACM CCS · NDSS — SoK tracks

WHAT IT CONTRIBUTES BEYOND A SURVEY

- New measurements, not re-quoted numbers — 16 cells that exist in no paper
- A threat-model result: defenses as attack amplifiers (0% → 99%), with the undefended control proving the attack is otherwise inert
- A falsifiable mechanism stated before the test and confirmed monotonically
- An actionable design rule, held across 2 families and 5 implementations
- Reproducibility evidence: 12 distinct defect classes documented with fixes

HONEST LIMITS, AND WHAT IS RUNNING NOW

- One retrieval dataset (open_nq); three models; four attack mechanisms
- The aggregation family comes from a single repository — a third-party implementation is the next real test
- A model-controlled rerun is executing now; its finished half already shows the KeywordAgg protection does not replicate
- Next: AGGD as a third reconstructable attack; close the 99 candidate missed pairs
- Then freeze the transfer matrix and write up

The through-line: a survey found that two communities never test each other's work. Running the code showed why that matters — several defenses are not merely untested against the other threat model, they are harmful under it.